

Program 7

Deploy a Web Application to Kubernetes using Minikube

DevOps

— Program7

— nginx-deployment.yaml

Step 1 : Start minikube and check the status of minikube

```
shreya@1RV24MC102-ShreyaC: ~/DevOps/Program7
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ minikube start
minikube v1.37.0 on Ubuntu 24.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: metrics-server, storage-provisioner, default-storageclass
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ minikube status
minikube
type: Control Plane
host: Running
kubenet: Running
apiserver: Running
kubeconfig: Configured
```

Step 2 : Add the following code in nginx-deployment.yaml file

```
GNU nano 7.2 nginx-deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 3 : Apply the nginx-deployment.yaml and check the status

```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ nano nginx-deployment.yaml
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
demo-nginx    1/1     1            1           29d
myapp         2/2     2            2           29d
nginx-app     1/1     1            1           29d
nginx-deployment 2/2     2            2           28s
web-app       1/1     1            1           29d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
demo-cron-29451744-46ggg            0/1     Completed 0           5m47s
demo-cron-29451746-r6gq9            0/1     Completed 0           3m47s
demo-cron-29451748-zz8x7            0/1     Completed 0           107s
demo-nginx-5c5b69c548-dm5np         1/1     Running   4 (13d ago) 29d
kubia                                1/1     Running   3 (7m10s ago) 22d
myapp-84dd7cdd78-25vcl              1/1     Running   4 (13d ago) 29d
myapp-84dd7cdd78-6765r              1/1     Running   4 (13d ago) 29d
nginx-app-66cf4d5595-nx9gm          1/1     Running   4 (13d ago) 29d
nginx-deployment-6f9664446b-5wgdk   1/1     Running   0           34s
nginx-deployment-6f9664446b-p4kk8   1/1     Running   0           34s
program-8                            1/1     Running   1 (7m10s ago) 13d
web-app-c8b4f854b-wp2hl             1/1     Running   4 (13d ago) 29d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$
```

Step 4 : Expose the deployment as a service

```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ kubectl expose deployment nginx-deployment \
--type=NodePort \
--port=80
service/nginx-deployment exposed
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$
```

Step 5 : Check the service and access the web app

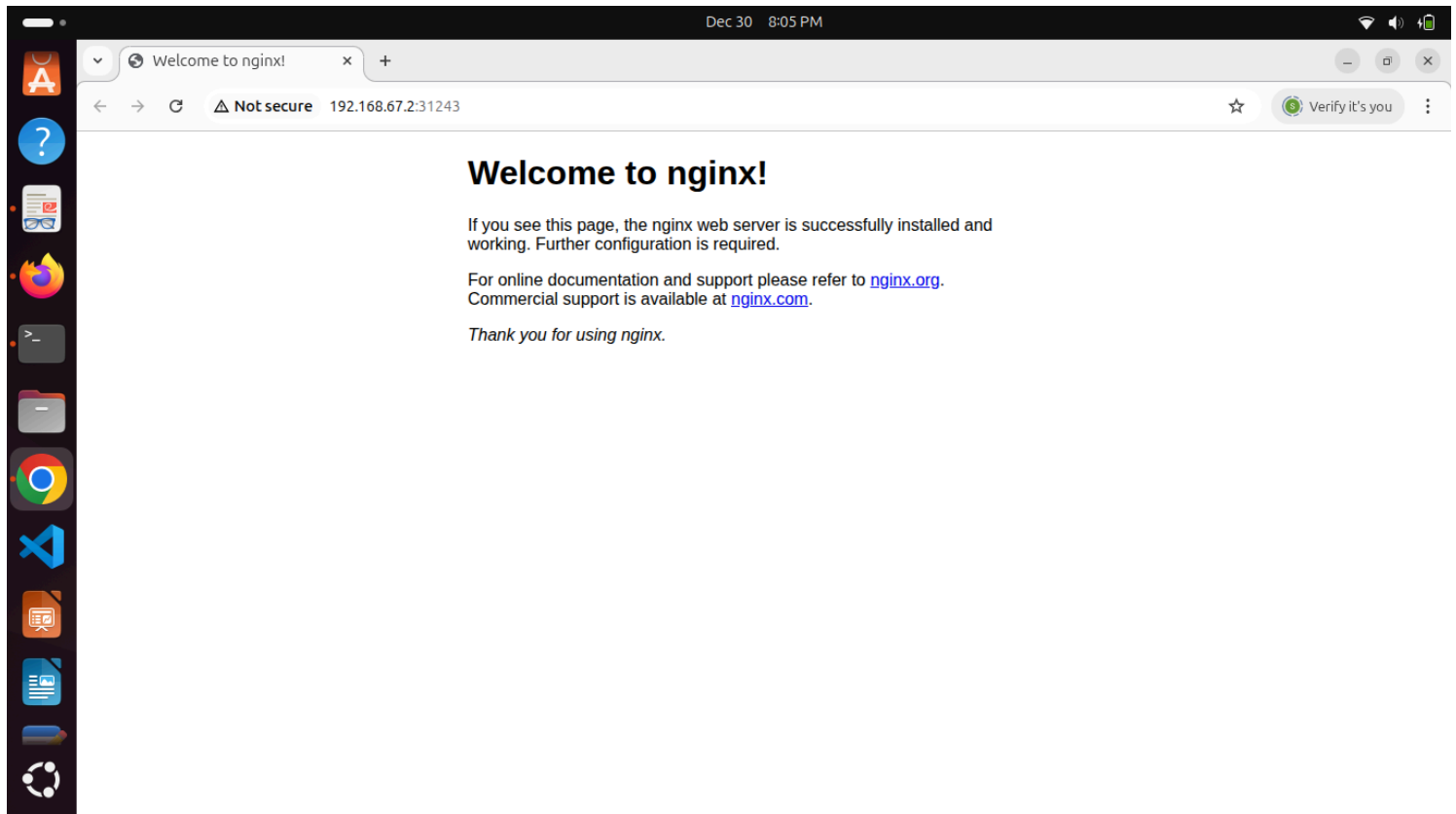
```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ kubectl get services
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes    ClusterIP   10.96.0.1    <none>        443/TCP          34d
kubia         NodePort    10.96.6.162  <none>        8080:30776/TCP   22d
nginx-deployment NodePort    10.103.54.102 <none>        80:31243/TCP     65s
test-pod      NodePort    10.97.158.48 <none>        80:32208/TCP     29d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ minikube service nginx-deployment

```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.67.2:31243

```

Opening service default/nginx-deployment in default browser...
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ Opening in existing browser session.
```



Step 6 : Clean-up process and delete all the services and deployments

```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ kubectl delete pods --all
pod "demo-cron-29451752-vmvw2" deleted from default namespace
pod "demo-cron-29451754-x2r7z" deleted from default namespace
pod "demo-cron-29451756-8pfc7" deleted from default namespace
pod "demo-nginx-5c5b69c548-dm5np" deleted from default namespace
pod "kubia" deleted from default namespace
pod "myapp-84dd7cdd78-25vcl" deleted from default namespace
pod "myapp-84dd7cdd78-6765r" deleted from default namespace
pod "nginx-app-66cf4d5595-nx9gm" deleted from default namespace
pod "nginx-deployment-6f9664446b-5wgdk" deleted from default namespace
pod "nginx-deployment-6f9664446b-p4kk8" deleted from default namespace
pod "program-8" deleted from default namespace
pod "web-app-c8b4f854b-wp2hl" deleted from default namespace
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
service "kubia" deleted from default namespace
service "nginx-deployment" deleted from default namespace
service "test-pod" deleted from default namespace
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ kubectl delete deployments --all
error: the server doesn't have a resource type "deploments"
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$ kubectl delete deployments --all
deployment.apps "demo-nginx" deleted from default namespace
deployment.apps "myapp" deleted from default namespace
deployment.apps "nginx-app" deleted from default namespace
deployment.apps "nginx-deployment" deleted from default namespace
deployment.apps "web-app" deleted from default namespace
shreya@1RV24MC102-ShreyaC:~/DevOps/Program7$
```